

Oh, the Places Formaldehyde Will Go!

(The Production of Formalin)

Homework No. 1
CHE 396 – Senior Design I
September 17, 2019

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Project Charter

Title	Goal	In Scope	Out of Scope
<i>Oh, the Places Formaldehyde Will Go!</i> (The Production of Formalin)	To analyze and select the optimal process for the production of Formaldehyde.	Design of Formalin production from methanol with economic considerations	- EPA Compliance - Location Impact, Raw Material Purchase, Packaging, and Distribution - Safety Consideration - Capital Investment
Deliverables		Team Members	
- Design Plan - Market Analysis - Solution Comparisons - Process Flow Diagram - Mass Balance		Evelyn Almonte Thomas Mazur Caleb Tristano Yohannes Teweldeberhan	

Design Plan

Task ID	Task	Start Date	Duration	Finish Date	Assignee
Logistics					All
1	First Meeting (Task Delegation)	9/10/19	1	9/10/19	All
2	Market Analysis	9/10/19	1	9/10/19	Yohannes/Caleb
3	Technology Comparison Research	9/12/19	5	9/16/19	All
4	BFD & PFD Research	9/13/19	2	9/17/19	Evelyn/ Yohannes
5	Project Charter				All
6	Design Plan	9/17/19	1	9/17/19	All
Execution					
Introduction					Evelyn
7	Objective	9/11/19	6	9/17/19	Evelyn/Thomas
8	Market Analysis	9/13/19	3	9/15/19	Yohannes/Caleb
Process Description					All
9	Problem	9/11/19	6	9/17/19	Thomas
10	Possible Solutions	9/11/19	6	9/17/19	Thomas
11	Chosen Solution	9/11/19	6	9/17/19	Thomas/Caleb
12	Design Basis and Assumptions	9/15/19	1	9/15/19	Caleb
13	BFD & Rough Mass Balance	9/13/19	5	9/17/19	Yohannes & Evelyn
14	PFD & Explanation	9/14/19	4	9/17/19	Evelyn

Wrap Up					
15	Layout/Format	9/11/19	6	9/17/19	Evelyn/Caleb
16	Section Unification	9/15/19	3	9/17/19	Caleb
17	Proofreading/Plagiarism Check	9/16/19	2	9/17/19	Caleb

All Indicates: Evelyn, Thomas, Caleb & Yohannes

Introduction

Objective

Formaldehyde Consulting, Inc. has been hired by Formalin Productions to develop a report regarding the most effective chemical synthesis of formaldehyde to be able to deliver Formalin to consumers using Formalin in their production of products such as adhesives, binders, preservatives, and other items. The aim of this report is to compare the different processes involved in the production of formaldehyde and detail an optimal method of production given resources and market forces. Formaldehyde has various names such as Methanal, Preservative, and E240. Because formaldehyde is most commonly sold as Formalin, this analysis initially started with the assumption that our product would be Formalin, which is an aqueous solution of 37% formaldehyde with 10-15% methyl alcohol in solution in order to prevent formaldehyde from polymerizing.ⁱ However, we deem it important to note, that it is actually only pure Formaldehyde that is analyzed, and that the additions of water and methanol to make the product transportable are known entities and do not need to be considered at this stage.

Market Analysis

Formaldehyde is used primarily to produce resins, as seen below:

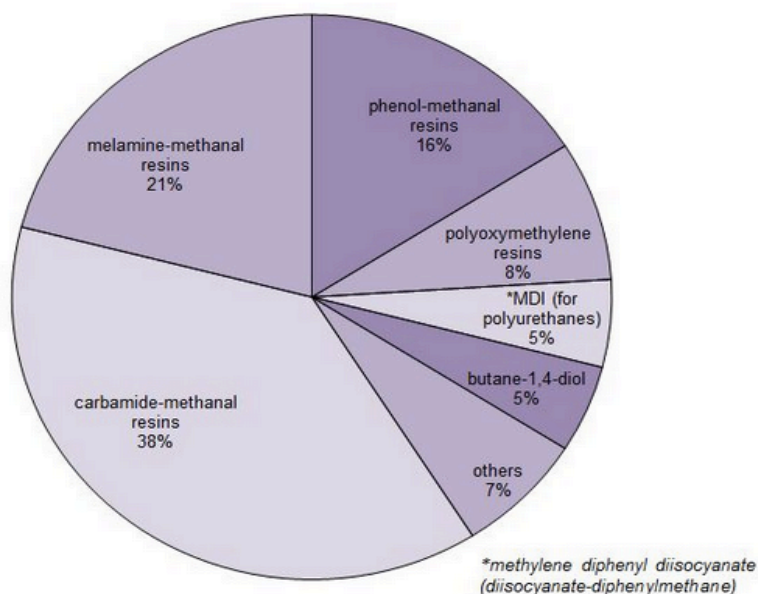


Figure 1: Uses of Methanal: Data for 2015, MMSA, 2016ⁱⁱ

“World consumption of 37% formaldehyde is forecast to grow at an average annual rate of about 2.5% from 2017 to 2022, similar to the rate of growth during 2012–17. Northeast Asia, especially China, continues to add capacity for formaldehyde, as the largest producer globally.”ⁱⁱⁱ

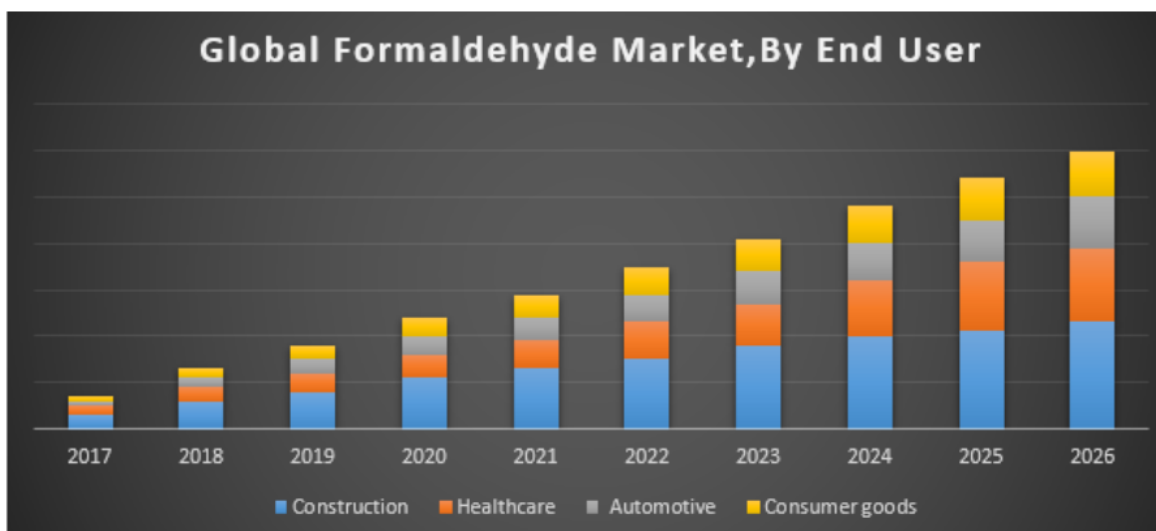


Figure 2: Global Formaldehyde Market in 2017^{iv}

Emerging economies, such as India and China, are expected to witness growth in the formaldehyde market. “The growing trend of antique furniture and wood based floors, walls, etc. is increasing the consumption of plywood, wood panel board products and particleboard”.^v This in turn, is leading to an increase in the consumption of adhesives and is thereby impacting the formaldehyde market positively as formaldehyde is a significant component of adhesives.^{vi} The increasing interest of people toward maintenance of houses and buildings is surging the consumption and application of paints and coatings, varnishes and enamels. Additionally, increasing population necessitates increased food production and in turn putting pressure on farmers, who are consuming higher amounts of fertilizers thus, fueling the market.^{vii}

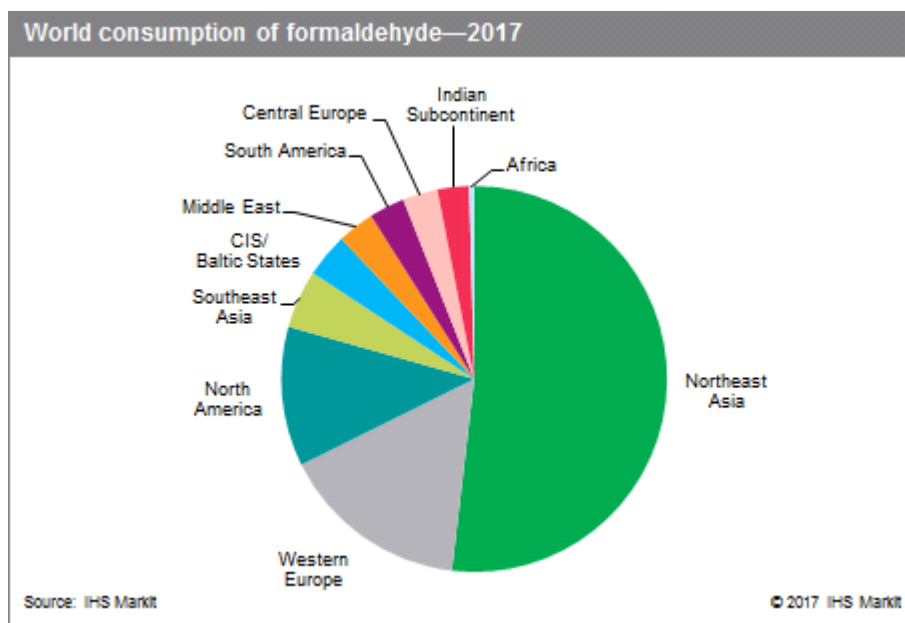


Figure 3: World Consumption of Formaldehyde 2017^{viii}

“North America and Europe are expected to witness significant growth in the market due to the presence of numerous drug and dye industries in the region”.^{ix} Also, the markets in Middle East and Latin America are expected to witness growth during the forecast period due to growth in the Latin America agricultural industry and the increasing number of construction projects in the Middle East and Africa region.

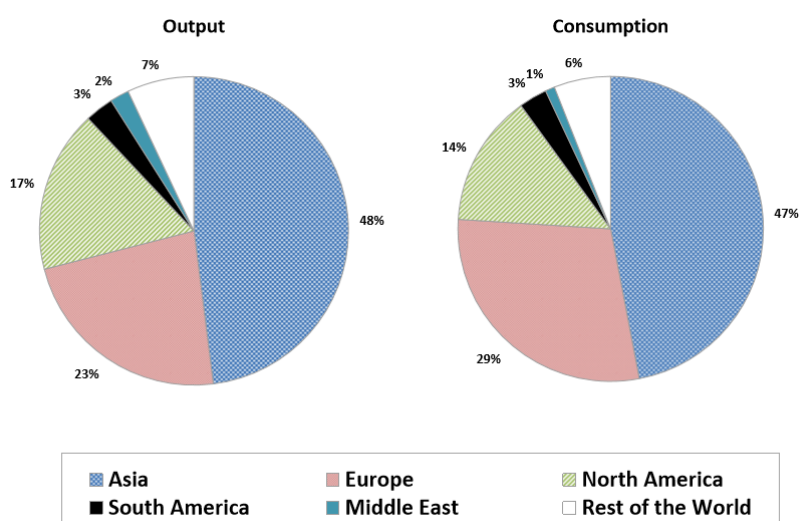


Figure 4: Output vs. Consumption by Geographic Region^x

Most formaldehyde producers are concerned primarily with satisfying captive requirements for derivative and/or supplying local merchant sales. “Formaldehyde is usually produced close to the point of consumption since it is fairly easy to make”, but it is costly to transport and can develop problems associated with stability during transport^{xi}. As a result, world trade in formaldehyde is minimal. “The formaldehyde market in North America has decreased by 15% since 2006”.^{xii} The significant reason for the decline has been the impact of the poor housing market and the construction industry. “According to Momentive Specialty Chemicals (formerly known as Hexion) in 2007, the formaldehyde market in North America was at just over 4 million tons and was roughly half of that of Europe”.^{xiii} Hence the formaldehyde demand will depend on economic recovery, especially in the housing and construction industries.

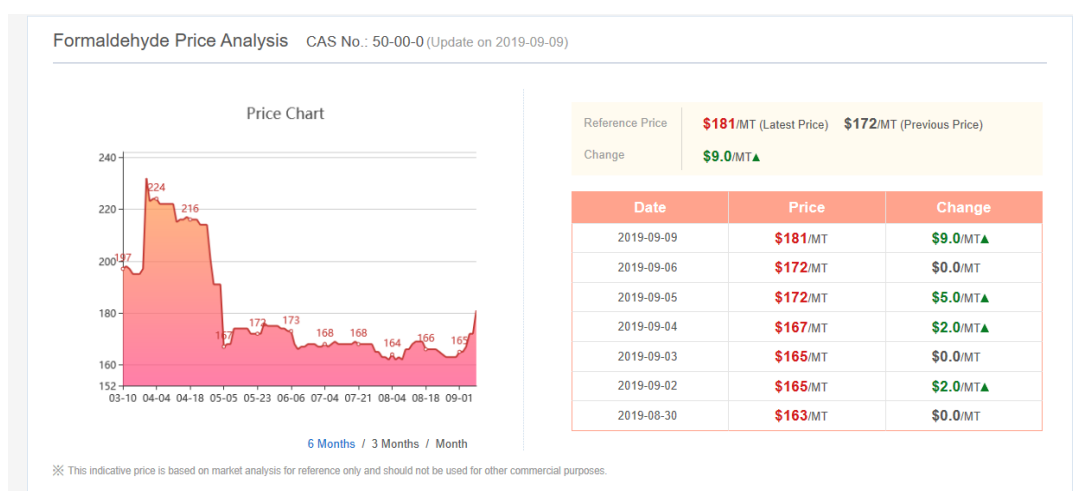


Figure 5: Formaldehyde Price For Six Month Period^{xiv}

With ever increasing awareness of health effects of Formaldehyde exposure, producers are coming under scrutiny from various government and non-governmental agencies. With continued reports of Formalin as a carcinogen, regulators throughout the world are instituting stringent measures to protect human health, consequently affecting the price of the product in unpredictable ways.

The point? Formaldehyde is used where it is made and the demand for the same is growing 

Process Description

Problem

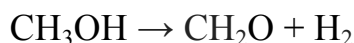
The production of formaldehyde generally requires one chemical reaction. That is, either the catalytic partial oxidation or dehydrogenation of methanol to produce formaldehyde and water. It is possible to produce formaldehyde using methane, however methanol is easier to oxidize.^{xv} Health concerns related to formaldehyde mainly involve the decomposition of resins and adhesives which can cause throat and eye irritation.^{xvi} Due to instabilities associated with formaldehyde, including polymerization and oxidation with air, formaldehyde/formalin are often not transported internationally and are typically produced close to consumers.^{xvii} The various methods used in the production of formaldehyde come with their own respective strengths and limitations. It is our challenge to identify which of these methods best suits the production of formaldehyde based on the concept that formaldehyde production cannot be globally centralized, but needs to be regional and readily available.

Possible Solutions

We can consider three options for the production formaldehyde:

1. Dehydrogenation of methanol,
2. Partial oxidation of methanol using a silver catalyst, or
3. Partial oxidation using various metal oxide catalysts.

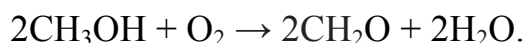
The dehydrogenation process involves the following endothermic reaction:



It is important to note that since we have an endothermic process, meaning heat must be

supplied to initiate the reaction. This process requires a high temperature in the approximate amount of 726.85°C to achieve a 98% conversion of methanol. A high conversion of methanol to formaldehyde is desired because if the conversion is not complete, any reactants left in the product will be difficult to separate. Even at an increased temperature to 759.85°C, Zaza and Renken note that a yield of 79% is achieved with the use of a silver catalyst. A major drawback of this process includes many competing reactions in the dehydrogenation reaction. Some of these competing products include CO, CH₄, H₂O, C, CH₃OCH₃, and HCOOCH₃^{xviii}. The required temperature range of dehydrogenation to achieve a desirable conversion ends up being greater than that of the next two processes and gives lower yields.

The partial oxidation processes of methanol involves the following exothermic reaction:



When using a manganese-molybdenum catalyst, the optimal temperature was determined to be around 365°C with yields varying between 72 to 83% by Mann and Hahn.^{xix} What is beneficial about this process is that there are less side products when compared to dehydrogenation. Mann and Hann list N₂, O₂, H₂O, and HCHO in a product analysis.^{xx} When using a silver catalyst as opposed to a metal oxide catalyst, endothermic dehydrogenation takes place as a side reaction.^{xxi} However, with the silver catalyst process, yields of 88 to 92% were achieved but required a higher temperature of around 600°C. It's important to note that a distillation column along with absorbers are required for the silver catalyst process for further purification which could add to our utilities cost.^{xxii}

Chosen Solution

Our chosen solution will be to produce formaldehyde via partial oxidation of methanol with a silver oxide catalyst reactor operating at approximately 600°C. This process was chosen over the other two methods after consideration of the pros and cons for each method.

Dehydrogenation of methanol was not selected because it is an endothermic reaction. As the heat must be supplied to it combined with the higher levels of temperature needed to achieve the desired output level range it is not economically feasible. The utility cost to produce the necessary heat are prohibitive.

After the elimination of the first possible solution, the options left both involve using a catalyst in partial oxidization. In terms of straight yields, silver catalyst yields are higher than those when compared to metal oxide.^{xxiii}

It must be noted that in terms of life span for the catalysts, metal oxide catalysts can last 12-18 months, versus silver catalysts that must be changed every 3 to 4 months. However, only a portion to a metal oxide catalyst can be reused, specifically the molybdenum. Alternatively, silver oxide catalyst can be regenerated.^{xxiv} Therefore, in the long run plants utilizing silver catalyst can realize significant savings from the regeneration of their catalyst.

As previously discussed, utility costs for the designation of a method are a continued concern. By choosing partial oxidation with silver catalyst, the reactor will heat to 600°C due to the exothermic nature of the reaction. This will allow a considerable amount of thermal energy to be recycled and distributed to other parts of the plant like the distillation column or preheating feed streams. This can significantly reduce the utility costs that would be incurred if the heat had to be generated from an external utility.

Additionally, to further optimize the plant we propose to use the hydrogen gas created as a bi-product of the reaction in fuel cells to produce electricity and pure water. The electricity can be used to reduce the facility's utility costs. The pure water can be used in the heat exchange system that removes heat from the reactor and transfers it to the distillation column and other components. Additionally, as discussed in the objective, pure water is one of the necessary components when turning formaldehyde into formalin for commercial distribution. In the long term, the economic potentials of using the silver catalyst method outweigh the drawbacks when looking at the operating costs of the plant as a whole rather than the singular process.

With this information in hand, we can take a region to illustrate. What if the decision was made to construct the new formalin manufacturing facility in China. The Chinese demand for formalin is expected to grow at approximately 2.6% almost indefinitely.^{xxv} Annual consumption for formalin was 6100 kt (kilotonnes) in 2004.^{xxvi}

With that in hand we can reasonably estimate 2019 production to be ~8965 kt and for demand to grow by over 233 kt between 2019 and 2020. Based on our plant design (silver oxide catalyst production) is isn't economically feasible to produce more than 100 kt/yr in our plant. The market value of formalin does experience wide fluctuation, but a conservative estimate of selling price for our formalin is 300 USD/t (US Dollars per tonne). At a production level of 90kt/yr, estimated gross income is ~ \$27,000,000 /yr.

Design Basis

Our design basis derives from an increase in Chinese demand for formaldehyde of approximately 2.6%/yr. We expect an increase of approximately 233 kt between 2019 and 2020, and forecasts suggest that this number will increase in subsequent years.^{xxvii} In order to help meet this increased demand we recommend designing a plant to produce 90kt/yr (or 90,000 tonnes/yr).

Operating Conditions, Flow Rates and Assumptions

1. Conversion of Methanol: 87.4%
2. Selectivity of Reaction: 99%
3. Formaldehyde Needed: 90,000 metric tons/year
4. Inlet Temperature From Reactor: 600C
5. Outlet Temperature From Reactor: 200C

BFD and Material Balance

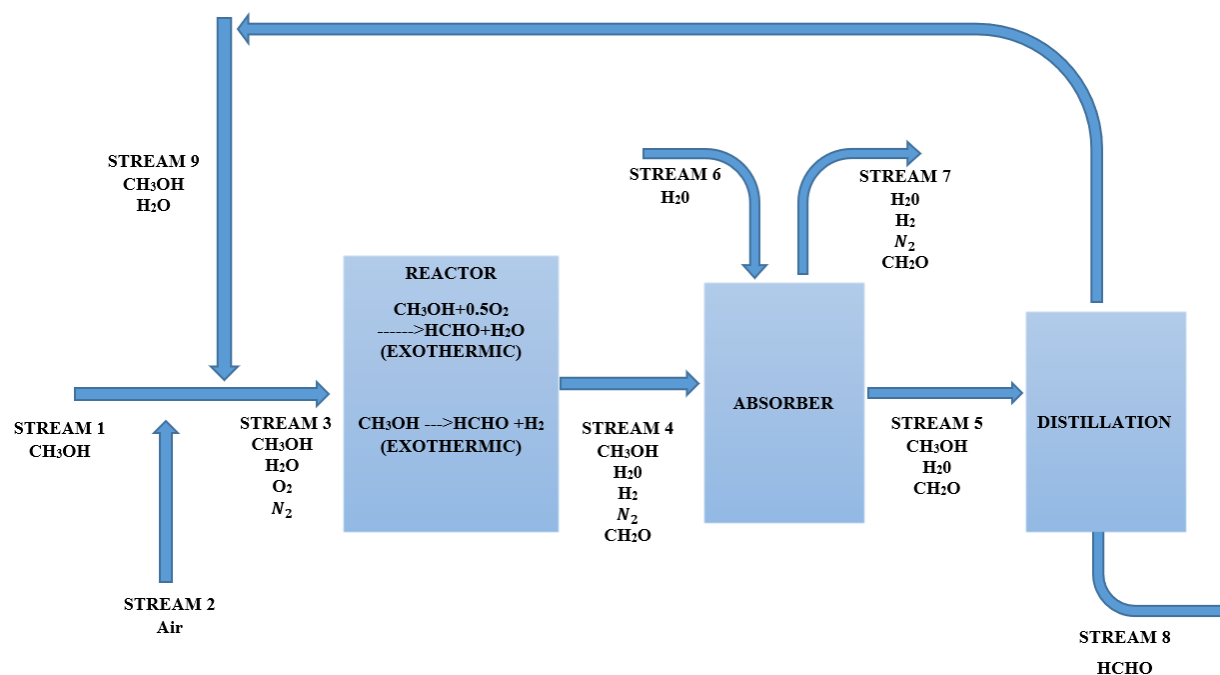
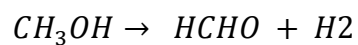
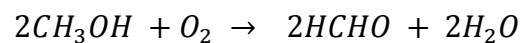
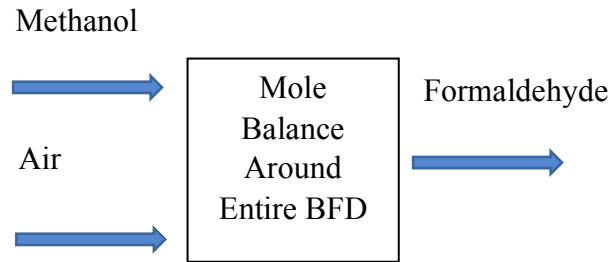


Figure 6: Block Flow Diagram of the Chemical Process

Rough Total Mass Balance:





$$\text{Goal: } \frac{378.459 \text{ kmol CH}_3\text{OH}}{\text{hr}}$$

$$\text{Methanol inlet: } \frac{378.459 \text{ kmol}}{\text{hr}} \text{ based on stoichiometry of 3:3}$$

$$\text{Air inlet: } \frac{126.153 \text{ kmol Air}}{\text{hr}} \text{ based on stoichiometry of 3:1 [formaldehyde: Air]}$$

	Molecular Weight	REACTOR(kmol/hr)		MASS BALANCE(gm/hr)	
		IN	OUT	IN	OUT
Methanol	32.042	437.3355985	55.10428542	14013107.25	1765651.513
Water	18.015	378.409	382.2313131	6817038.135	6885897.106
Oxygen	31.999	218.6677993	193901.0453	6997150.909	6059597.028
Formalin	30.03	0	382.2313131	0	11478406.33
Hydrogen	2.016	0	812.3731707	0	1637744.312
		Sum		27827296.29	27827296.29
		Difference		3.4932E-05	

	Molecular Weight	ABSORBER(kmol/hr)			MASS BALANCE(gm/hr)		
		IN	OUT(TOP)	OUT(BOTTOM)	IN	OUT(TOP)	OUT(BOTTOM)
Methanol	32.042	55.1042854	0.55104285	54.55324256	1765652	17656.51513	1747994.998
Water	18.015	382.231313	3.82231313	378.409	6885897	68858.97106	6817038.135
Oxygen	31.999	193901.045	193901.045	0	6.2E+09	6204639548	0
Formalin	30.03	382.231313	3.82231313	378.409	11478406	114784.0633	11363622.27
Hydrogen	2.016	812.373171	812.373171	0	1637744	1637744.312	0
		Sum			6.23E+09	6206478592	19928655.4
		Difference			0		

	Molecular Weight	DISTILLATION (Kmol/hr)			MASS BALANCE(gm/hr)		
		IN	OUT(TOP)	OUT (BOTTOM)	IN	OUT(TOP)	OUT(BOTTOM)
Methanol	32.042	54.55324256	54.55324256	0	1747994.998	1747994.998	0
Water	18.015	378.409	378.409	0	6817038.135	6817038.135	0
Oxygen	31.999	0	0	0	0	0	0
Formalin	30.03	378.409	0	378.409	11363622.27	0	11363622.27
Hydrogen	2.016	0	0	0	0	0	0
		Sum			19928655.4	8565033.133	11363622.27
		Difference			0		

Figure 7: Mass Balance Tables After Assumptions Outlined Below

Assumptions for Mass Balance:

Distillation Column

1. The amount of formaldehyde that is exiting the distillation column is pure.
2. In the calculation, we considered the formation of Formalin by adding deionized water to make the composition of 37% Formaldehyde. This extra step does not need to be done if

one would just like to obtain the pure Formaldehyde, but is rather incorporated to explore another viable product.

Absorber Column

1. For the absorber 99% absorption rate is used for formaldehyde.
2. Oxygen and hydrogen are completely removed.

Reactor

1. It has 87.4% conversion rate.
2. No Hydrogen is going into reactor.

PFD and Equipment Description

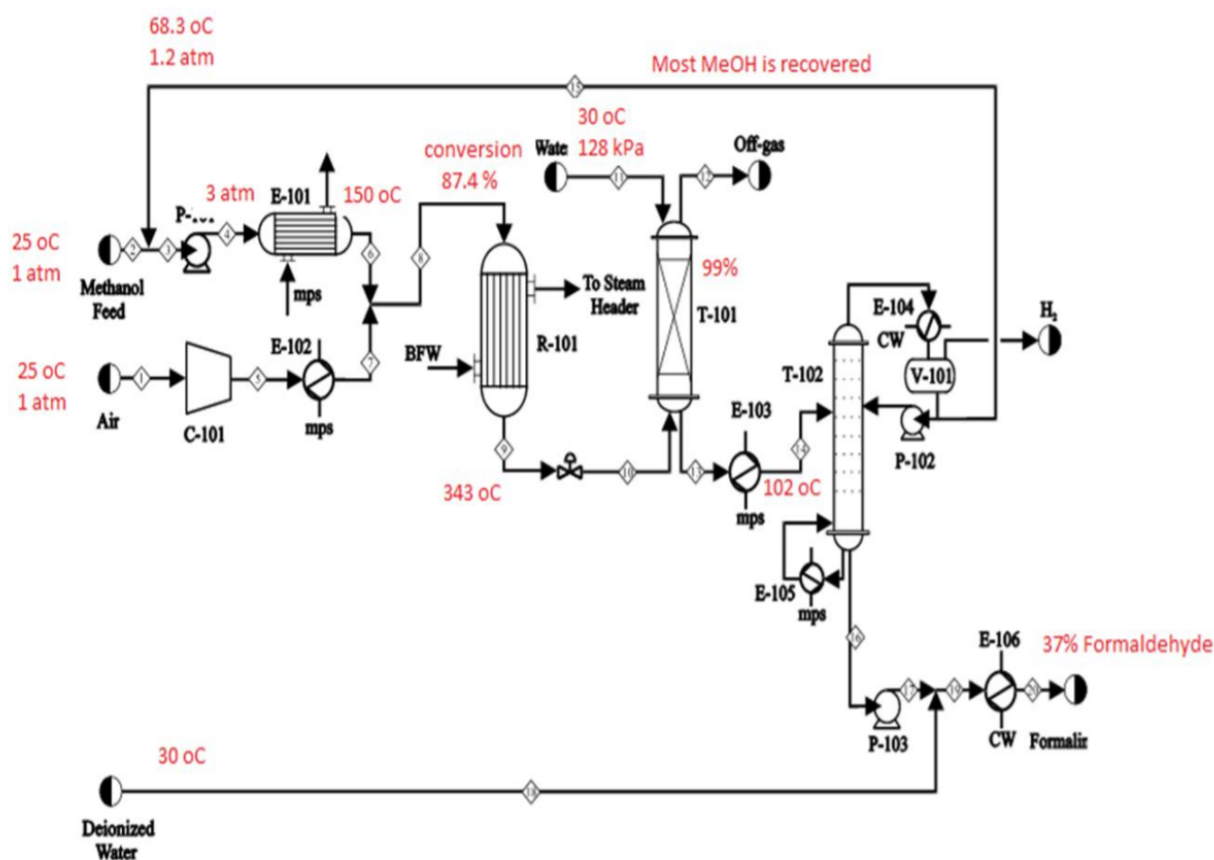


Figure 8: Process Flow Diagram of a Formaldehyde Plant^{xxviii}

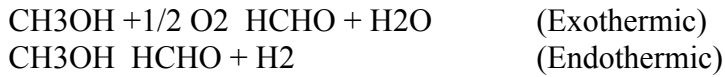
Design Equipment	Designation
Reactor	R-101
Absorber	T-101
Distillation column	T-102
Methanol heater	E-101
Air heater	E-102
Absorber effluent heater	E-103
Distillation condenser	E-104
Distillation reboiler	E-105
Formalin cooler	E-106
Air compressor	C-101
Methanol feed and recycle pump	P-101
Distillation bottom product pump	P-103
Fresh air line	stream 1
Fresh methanol line	stream 2
Fresh methanol and recycle line	stream 3
Methanol line pumped by P-101	stream 4
Compressed Air line by C-101	stream 5
Methanol line heated by E-101	stream 6
Air line heated by E-102	stream 7
Mixing line of methanol and air	stream 8
Reactor effluent	stream 9
Absorber inlet line	stream 10
Fresh water inlet to absorber	stream 11
Absorber off gas line	stream 12
Absorber effluent	stream 13
Heated distillation tower inlet by E-103	stream 14
Distillation top recycle line	stream 15
Distillation bottom line	stream 16
Pumped distillation bottom product by P-103	stream 17
Dilution deionized water line	stream 18
Water and formaldehyde mixing line	stream 19
Cooled formalin product by E-106	stream 20

Figure 9: Design Equipment Continuation of Figure 8^{xxix}

Equipment Descriptions

1. **Compressor:** a machine used to supply fluid at an increased pressure.
 - a. C-101: Assumption: $N=0.75$; Adiabatic; Constant heat capacities.; Ideal gas.
2. **Reactor:** causes a reaction that converts feed into desirable material (products).
 - a. a)R-101

The reactions involved are:



3. **Absorber:** brings gas and liquid phases in contact, so that contaminants in the gas phase are absorbed into the liquid phase as a result of their interaction.
 - a. T-101 packed-bed absorber used for gas removal and assuming the column is packed with 2" Ceramic Berl Saddle.
4. **Distillation Column:** this separates a compound having two or more mutually soluble solution based on the boiling point.
 - a. T-102 22 stages with a 10 m spacing
5. **Heater:** A device that is used to transfer heat between one or more fluids.

Assumed tube diameter = 0.012 m
Assumed wall thickness = 0.001 m

- a. Methanol Heater (E-101)
Assumed tube length = 1.1 m
 - b. Air Heater (E-102)
Assumed tube length = 1.3 m
 - c. Absorber Effluent Heater (E-103)
Assumed tube length = 1 m
6. **Condenser:** a device that is used to condense gaseous substance back into its liquid form through cooling
 - a. Distillation Condenser (E-104)
7. **Reboiler:** heat exchangers typically used to provide heat to the bottom of industrial distillation columns.
 - a. Distillation Reboiler (E-105)
8. **Cooler:** Used to cool down the product
 - a. Formalin Cooler (E-106)
9. **Pump:** a device used to move fluids by mechanical action and change in pressure.
 - a. Methanol feed and recycle pump (P-101)
 - b. Distillation bottom product pump (P-103)

References

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- ^{iv} “Global Formaldehyde Market - Global Industry Analysis and Forecast (2018-2026).” *MAXIMIZE MARKET RESEARCH*, www.maximizemarketresearch.com/market-report/global-formaldehyde-market/23567/.
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- ^{vi} Formalin Market Analysis
- ^{vii} Formalin Market Analysis
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^{xxii} Sanhoob, 2012

^{xxiii} Sanhoob, 2012

^{xxiv} Sanhoob, 2012

^{xxv} Formaldehyde, CHE Markit

^{xxvi} Tang, Xiaojiang, et al. *Formaldehyde in China: Production, Consumption, Exposure Levels, and Health Effects*. Environmental International, Nov. 2009, www.researchgate.net/publication/222421215_Formaldehyde_in_China_Production_consumption_exposure_levels_and_health_effects.

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^{xxviii} Sanhoob, 2012

^{xxix} Sanhoob, 2012